



Invited Talk

Linguistic Ontologies and Wordnets

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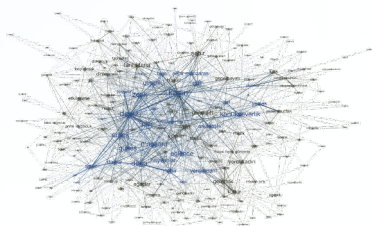
A man with glasses and a dark jacket is smiling while holding a large stack of books in a library. The shelves behind him are filled with various books, including sets of encyclopedias and dictionaries. The text 'Building linguistic resources for NLP' is overlaid on the right side of the image.

Building linguistic resources for NLP



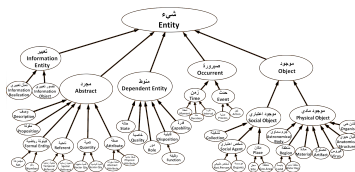
Lexical Resources at Birzeit University

Lexicographic Database



150 lexicons
Very large Arabic-
multilingual database

Arabic Ontology



Formal Arabic Wordnet
with ontologically clean
content

Dialect Corpora



Annotated corpora
each word is annotated
with many morph features

Big Linguistic Data Graph

<https://ontology.birzeit.edu>

Outline

- Why linguistic ontologies
- Wordnet Vs linguistic ontology Vs application ontology
- The Arabic Ontology

- ❖ Wordnets are used to understand and retrieve **unstructured information** in e.g., NLP and IR.
- ❖ New demands to use wordnets like ontologies: to manage and retrieve **structured data** in e.g., Knowledge Graphs, multilingual Big Data, and medical informatics.

Google Scholar

wordnet AND "knowledge graph"

Articles

About 4,770 results (0.05 sec)

Any time

Since 2021

Since 2020

Since 2017

Custom range...

Sort by relevance

Sort by date

☒ include patents

☒ include citations

☒ Create alert

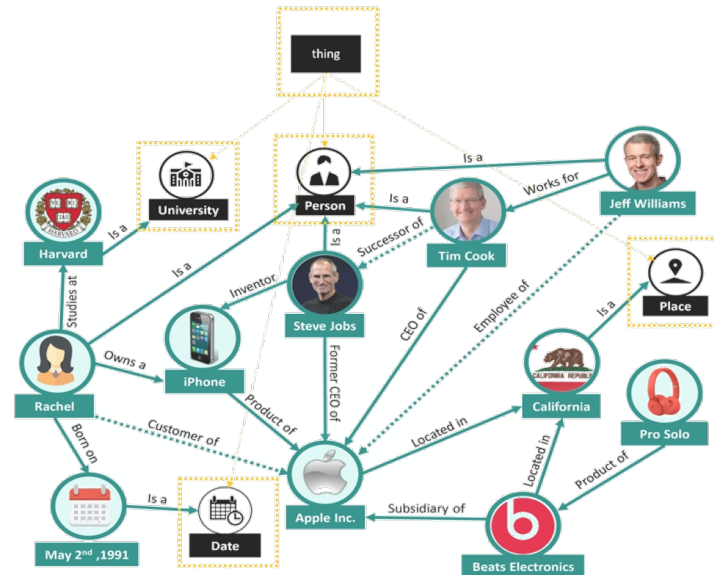
Knowledge graph embedding by translating on hyperplanes
 Z Wang, J Zhang, J Feng, Z Chen - ... of the AAAI Conference on Artificial Intelligence, 2014 - ojs.aaai.org
 ... predict h given (r, t). Rather than requiring one best answer, this task emphasizes more on ranking a set of candidate entities from the **knowledge graph**. We use the same two data sets which are used in TransE (Bordes et al. 2011; 2013b): WN18, a subset of **Wordnet**; FB15k, a ...
 ☆ 99 Cited by 1420 Related articles All 8 versions

Unsupervised word-level affect analysis and propagation in a lexical knowledge graph
 M Fares, A Moufarrej, E Jreij, J Tekli - Knowledge-Based Systems, 2019 - Elsevier
 ... unsupervised word-level **knowledge graph**-based LSA framework. It uses different variants of shortest path graph navigation techniques to compute and propagate affective scores in a lexical-affective graph (LAG), created by connecting a typical lexical KB like **WordNet**, with a ...
 ☆ 99 Cited by 11 Related articles All 2 versions

Learning entity and relation embeddings for knowledge graph completion
 Y Lin, Z Liu, M Sun, Y Liu, X Zhu - ... of the AAAI Conference on Artificial Intelligence, 2015 - ojs.aaai.org
 ... For this reason, traditional approach of link prediction is not capable for **knowledge graph** completion ... We evaluate our models with the tasks of link prediction, triple classification and relation fact extraction on bench- mark datasets of **WordNet** and Freebase ...
 ☆ 99 Cited by 1542 Related articles All 16 versions

A Novel Word Sense Disambiguation Approach Using WordNet Knowledge Graph
 M AlMousa, R Benlamri, R Khoury - arXiv preprint arXiv:2101.02875, 2021 - arxiv.org

- ❖ Wordnets are used to understand and retrieve **unstructured information** in e.g., NLP and IR.
- ❖ New demands to use wordnets like ontologies: to manage and retrieve **structured data** in e.g., Knowledge Graphs, multilingual Big Data, and medical informatics.



But,

Ontologies are typically application-specific rich axiomatizations;
Wordnets are general-purpose mental lexicons, and thus
axiomatizing wordnet would be a **rigidification**.

- How to build a linguistic ontology as a wordnet - to better serve new application scenarios

Wordnets Vs Ontologies



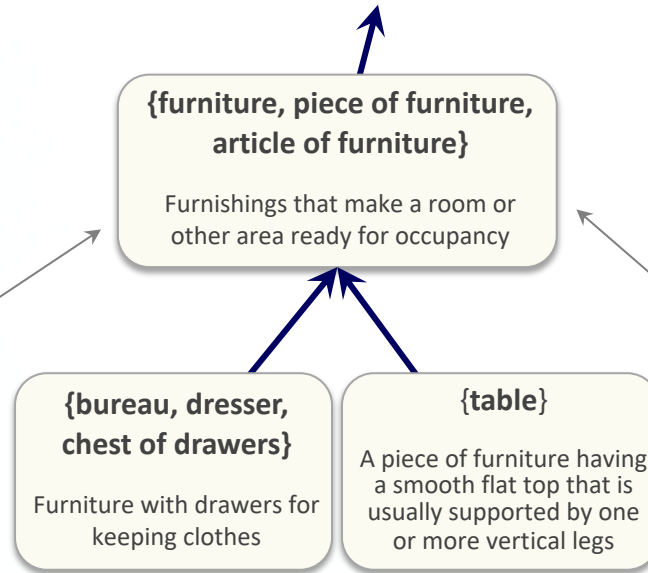
Synset vs Concept

Wordnet

made of synsets
(linguistic concepts)

Synset: *signifies a concept;*
a thought in our mind.

Individuals are also linguistic
concepts (ISO 1087-1:2000)



Ontology

made of concepts
(classes of individuals).

Concept: *class of individuals;*
characteristics its instances
have in common.

In BFO, Universal/Defined Class.

Proposed definition (for linguistic ontologies)

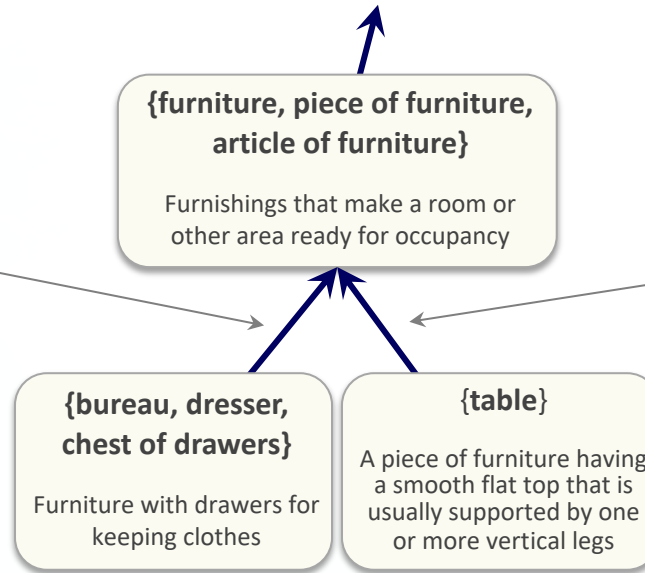
Definition: concept (Jarrar, 2021):

Given a concept c , its **intensional** interpretation c^I is defined on a *domain space* $\langle D, W \rangle$ as a function $c^I: W \rightarrow 2^D$, where D is a domain and W is a **set of maximal states of affairs** on D . For a concept c , the set $E_c = \{c^I(w) \mid w \in W\}$ is the set of the admissible extensions of c . Two concepts having the same set of *admissible* instances, in all states of affairs, are considered the same concept.

Hyponymy vs Subsumption

Wordnet

Hyponymy: If native speakers accept a sentence like: B is a kind of A



Ontology

Subsumption: a subset relation between concepts. Every instance in c_1 is also an instance in c_2 .

Formally: c_1 subsumes c_2 , iff every instance of c_2 is an instance of c_1 , in every possible state of affairs.

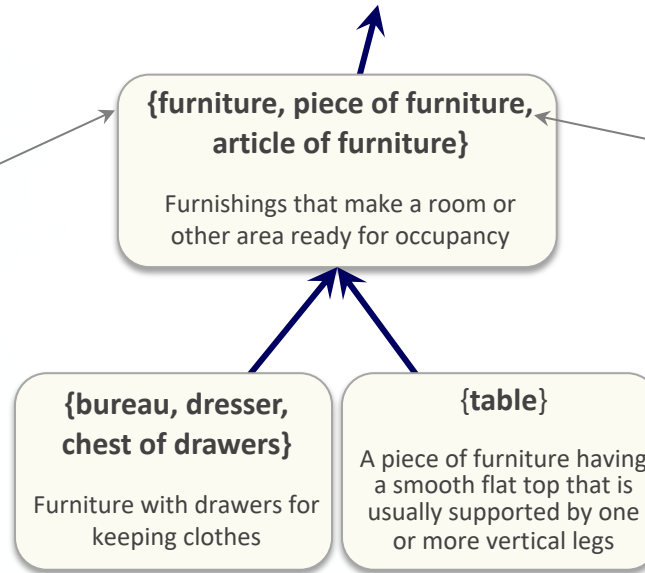
Synonymy

Wordnet

Synonyms: “two expressions are synonymous in a linguistic context C if the substitution of one for the other in C does not alter the truth value” (Miller et al., 1990).

Ontology

Synonyms: alternative labels/names of concepts.



Proposed definition (for linguistic ontologies)

Definition: Synonymy Relation (Jarrar, 2021)

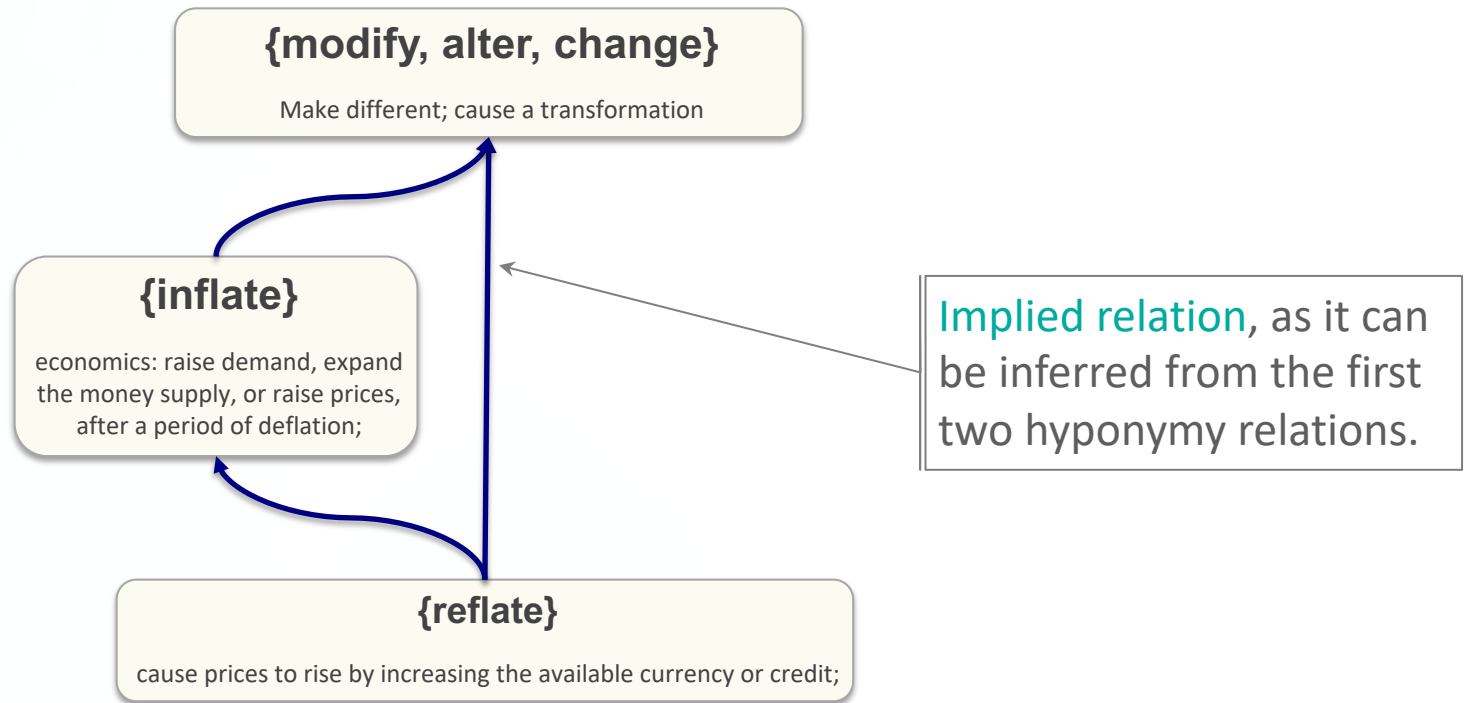
Given two terms t_1 and t_2 lexicalizing concepts c_1 and c_2 , respectively, then t_1 and t_2 are considered to be synonymous *iff* $c_1 = c_2$. In this way, synonymy can be defined as an **equivalence relation** $=_c$ **between terms** lexicalizing the same concept, thus it is a **reflexive, symmetric and transitive** relation.

More ontology/formal issues in wordnet:

Which might not be correct from a
formal/ontological perspective

Formal/Ontological issues in wordnet

No benefits for including implied relations



Formal/Ontological issues in wordnet

Might be different linguistic concepts, but ontologically it is the same instance.

{evening star, Hesperus, Vesper}

a planet (usually Venus) seen at sunset in the western sky

{Phosphorus, Lucifer, daystar, morning star}

a planet (usually Venus) seen just before sunrise in the eastern sky

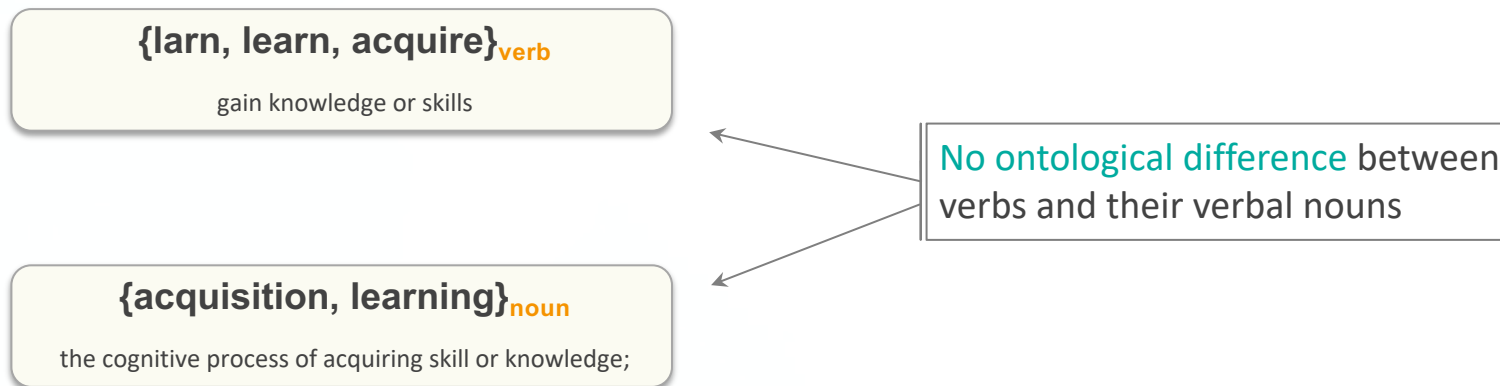
Same instance (i.e., Venus)
that people see at different
occasions.

Formal/Ontological issues in wordnet

Verbs are linguistic rather than ontological categories.

Ontologies capture the events that verbs denote rather than verbs themselves.

→ We say (he learns, he learned, he is learning, the learning he ..,) referring to the exact same learning event.



Formal/Ontological issues in wordnet

Who/How to decide on the accuracy?

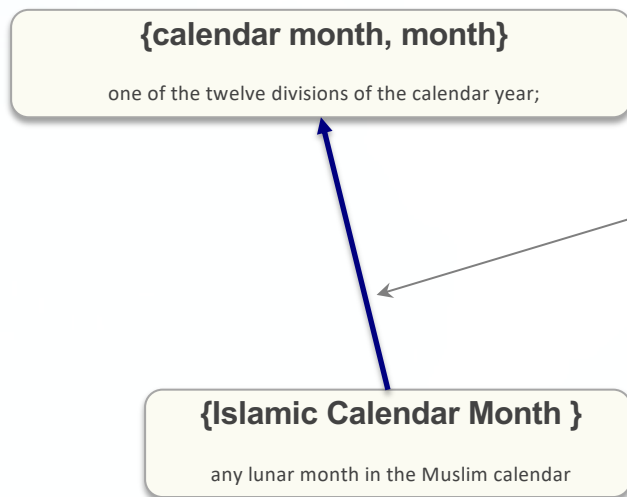
**{complex number, imaginary number,
complex quantity, imaginary}**

(mathematics) a number of the form $a+bi$ where a and b are real numbers and i is the square root of -1

not synonyms, Imaginary number is only a special case of a Complex number. Similarly, WordNet provides a poor classification of the types of numbers, e.g., Real, Rational, Natural, and Integer numbers are all subsumed by Number, while they subsume each other.

Formal/Ontological issues in wordnet

Who/How to decide on the accuracy?



Inaccurate, because Month is defined as one of the twelve divisions of the *calendar year*, which is a Gregorian year as defined in the gloss (i.e., an average month is about 30.43 days); however, an average of an Islamic month is 29.53 days. In fact, both months belong to two different calendar systems. A Gregorian month is 1/12 of a Gregorian year, while an Islamic year is a multiple of 12 lunar months.

Application Ontology vs Linguistic Ontology / Formal Wordnet

- Typically **rich** axiomatization
 - Each term refers to one concept (**no polysemy**).
 - Synonymy is not a target.
 - Benchmarked to **application's knowledge**.
 - Used by a **certain application** or a class of applications.
- **Light-weight** axiomatization, and cannot be rigid.
 - Each term refers to one or more concepts (**Polysemy**).
 - Synonymy is important.
 - Benchmarked **general knowledge**
 - Used for **general purposes**.

The Arabic Ontology

Arabic Ontology

- Characterization of the intended meaning (i.e., concepts) that the Arabic terms convey.
- **Formal Arabic Wordnet** - with ontologically-clean content.
- Linked with PWN, Wikidata, BFO, DOLCE
- Linked with many lexicons

The screenshot displays the Arabic Ontology web application. At the top, there is a search bar and navigation tabs for 'Ontology', 'Dictionaries', and 'Morphology'. The main content area shows the concept 'Entity' (شيء) with its Arabic and English definitions. Below this, there are four related concepts: 'Object' (موجود), 'Occurrent' (متبررة), 'Dependent Entity' (منوط), and 'Abstract' (مجرد). Each concept is accompanied by its Arabic name, English description, and an example sentence in Arabic. The interface also includes a footer with the Birzeit University logo and copyright information.

Entity | شيء | كَيْلُونَة | كائن
Whatever existed or will exist, and can be realized or imagined
أيما وُجد أو سيوجد ونستطيع إدراكه أو تخيلته
example: كل شيء على ما يرام
293198

Object | موجود | كائن | قائم | حقيقي | واقعي | شيء | ذات | قِيوم
An entity that is wholly and independently present in time, and is realized either for its concrete or social existence
شيء له ذات مستقلة بنفسه، وحاضر كلياً في الزمن، ويدرك بذاته قياساً أو لذاته اعتباراً
يختلف إدراكنا لأي موجود لاختلاف ما يميز أنواعه من الصفات الجوهرية
example: 293200 TypeOf: {Entity}

Occurrent | متبررة | حدوث | حصول
An entity realized by the time of its happening
الشيء الذي شارك ذاته وأجزائه بجريانه عبر الزمن
لا يمكن فهم أي حدث بشكل منفصل عن الإطار الزمني له
example: 293202 TypeOf: {Entity}

Dependent Entity | منوط | متعلّق | مشروط
An entity realized by the time of its happening
شيء يعتمد وجوده على وجود أشياء أخرى
طول المبنى منوط بوجود المبنى وإلا فلا طول له
example: 293201 TypeOf: {Entity}

Abstract | مجرد | تجريدي | غير مادي | نظري
An entity exists only in mind, cannot be measured or socially realized, and

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<https://ontology.birzeit.edu/concept/293198>

Arabic Ontology

- **Current size** so far (but the numbers are dynamic)
1800 fully-done concepts (mostly top levels)
17K partially investigated (ready for NLP applications)
Some branches are elaborated, other not yet.
- **English labels** are not our target - provided for readability and communication.
- **Methodology:** Built top-down and bottom-up at the same time.

▲ **occurent** | سُيَرورة | خُذوث | حُصُول
An entity realized by the time of its happening
الشيء الذي تُدرَك ذاته وأجزائه بجريانه عبر الزمن
لا يمكن فهم أي حدوث بشكل منفصل عن الإطار الزمني له
example: 293202 TypeOf: {entity}

▲ **process** | عَمَلِيَّة
A cumulative occurrent that is composed of a sequence of actions happening respectively in time
حدث تراكمي يتكون من سلسلة من الأفعال المترابطة، التي تحدث بشكل متتابع على خط الزمن
example: 293215 TypeOf: {occurent}

▲ **biological process** | single-organism process | single organism process | physiological process
عملية بيولوجية
A biological process represents a specific objective that the organism is genetically programmed to achieve. Biological processes are often described by their outcome or ending state
See More..
45500019 TypeOf: {process} Ontologies

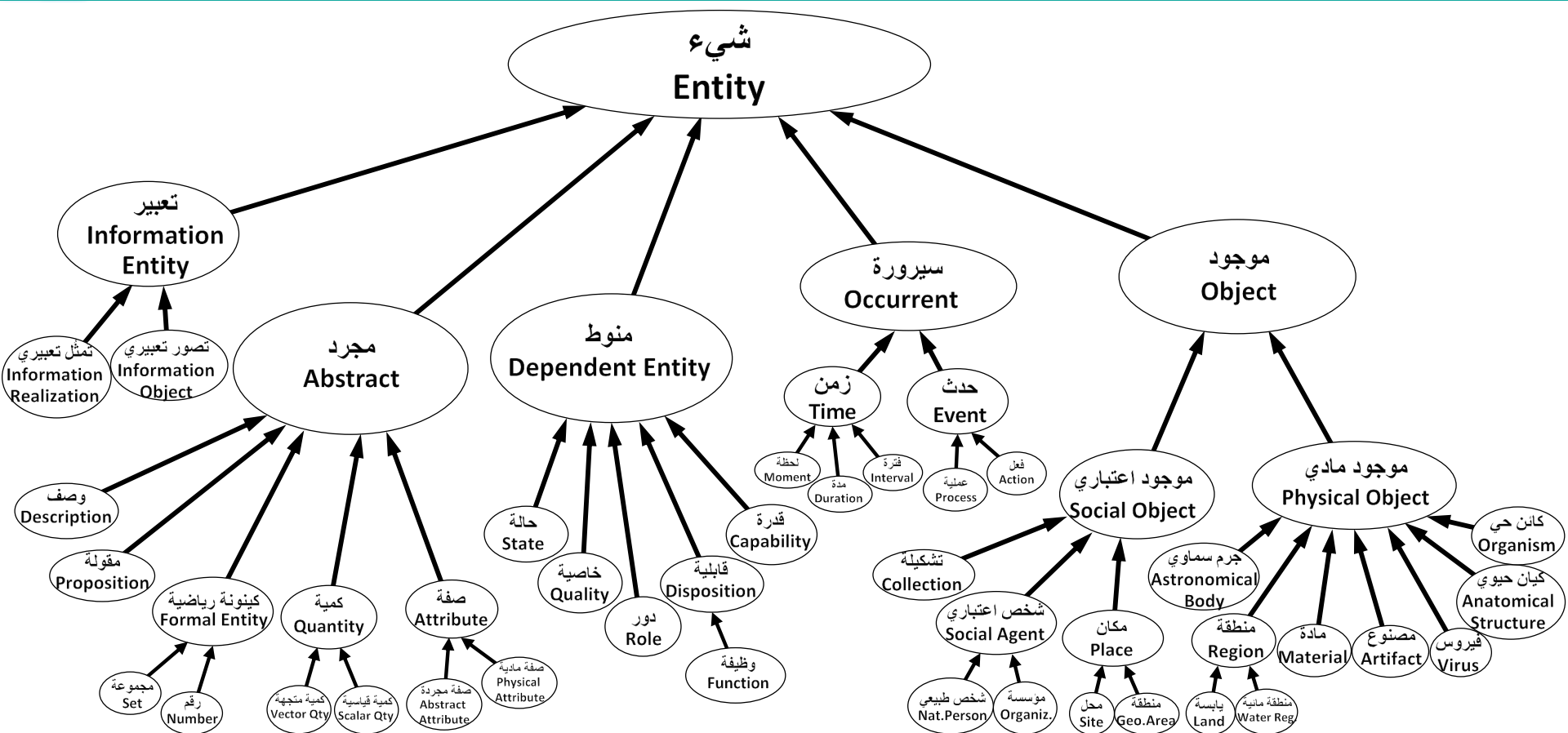
▲ **bodily process** | عملية جسمانية
A process in which at least one bodily component of an organism participates. | [OGMS_0000060]
455000574 TypeOf: {biological process} Ontologies

▶ **behavior** | سُلُوك
The internally coordinated responses (actions or inactions) of animals (Individuals or groups) to internal or external stimuli, via a mechanism that involves nervous system activity. [GO_0007610]
455000020 TypeOf: {bodily process} Ontologies

▶ **mental process** | عملية إدراكية
A mental process is a bodily process that is of a type

<https://ontology.birzeit.edu/concept/293198>

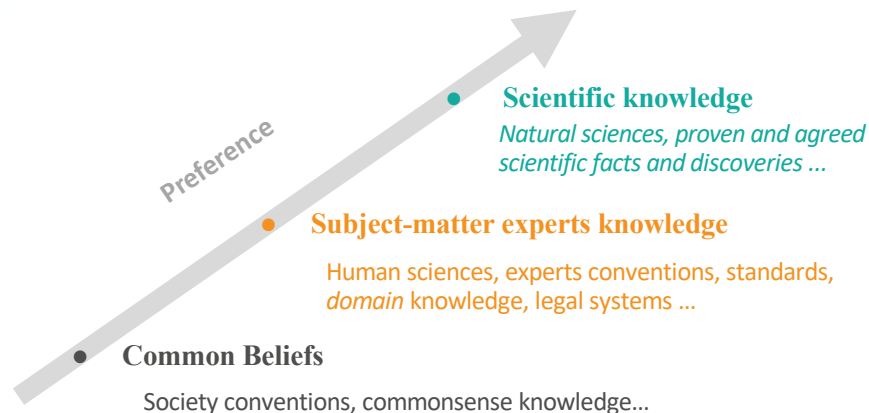
Top Levels of the Arabic Ontology



Benchmarking the ontology content

- What should the ontology capture and adhere to?
 - On what basis the correctness of the ontology content can be benchmarked?
-
- Should concepts be defined/classified based on what Arabic speakers commonly believe!
 - Should we adopt a certain lexicon and formalize it!
 - Should we rely on what the scientific literature accepts!
 - Should we build the ontology based on what we, the ontology builders, believe!

Benchmarking Methodology



Benchmarked against the following, in order:

1. **Scientific knowledge**, which scientists typically accept on the basis of experimentation and verification and commonly agree about. If no mature answer is found in the state-of-art scientific discoveries, then against,
2. **Subject-matter experts' and domain knowledge and conventions**. If no answer can be synthesized or attained from experts' knowledge, then against,
3. **Commonsense knowledge**, repeatedly found in quality lexicons and literature.

Gloss Formulation Methodology

The purpose of a gloss state the critical and distinguishing characteristics that all instances of a concept have in common, in an informal but controlled way as the following:

Step 1: Start with the supertype of the concept being defined. e.g., “Object: An entity that...”, “Physical Object: An object that ...”.

Entity شَيْءٌ | كَيْفُونَةٌ | كَائِنٌ
Whatever existed or will exist, and can be realized or imagined
أَيُّمَا وُجِدَ أَوْ سَيُوجَدُ وَنَسْتَطِيعُ إِدْرَاكَهُ أَوْ تَخَيُّلَهُ
example: كُلُّ شَيْءٍ عَلَى مَا يَرَامُ
293198

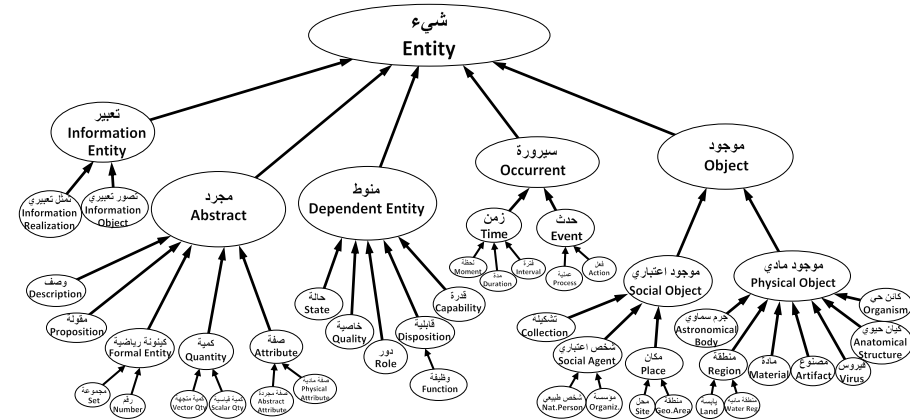
Object مَوْجُودٌ | كَائِنٌ | قائِمٌ | حَقِيقِيٌّ | وَاقِعِيٌّ | شَيْءٌ | ذَاتٌ | قَيُّومٌ
An entity that is wholly and independently present in time, and is realized either for its concrete or social existence
شَيْءٌ لَهُ ذَاتٌ مُسْتَقَلَّةٌ بِنَفْسِهِ، وَحَاضِرٌ كُلِّيًّا فِي الزَّمَنِ، وَيُدْرِكُ بِذَاتِهِ قِيَاسًا أَوْ لِذَاتِهِ اعْتِبَارًا
example: يَخْتَلِفُ إِدْرَاكُنَا لِأَيِّ مَوْجُودٍ لاختلاف ما يميِّز أنواعه من الصفات الجوهرية
293200 TypeOf : {Entity}

Step 2: List only the most distinguishing and intrinsic characteristics that specialize the concept from its supertype, and that differentiate it from other concepts in the same level.

Step 3: Write the distinguishing characteristics in the form of a sequence of propositions to help the reader to easily mentally rebuild the concept being defined in a declarative and non-narrative manner.

Comprehensiveness Evaluation

How much is the current version of the ontology able to *top* (i.e., be a supertype of) the concepts of the Arabic terms?



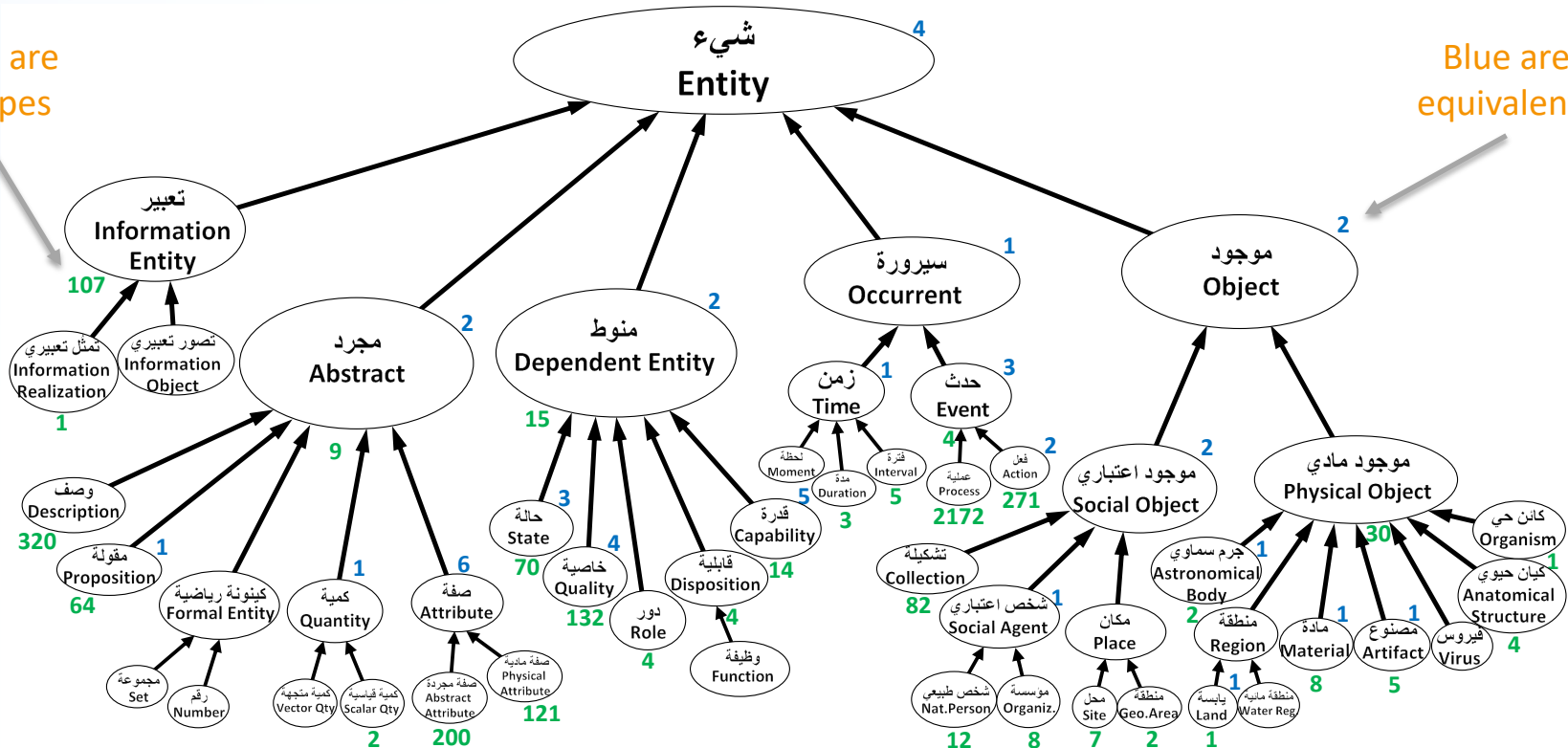
Experiment: classify the 1830 terms found in Al-Jurjānī “Ta‘rīfāt” lexicon (1339–1414 AD) under the ontology. This lexicon contains the most abstract notions in Arabic in most domains. 270 concepts are excluded (not understand, can be instantiated , etc).

Ideally: each of the 1830 concepts should be placed either as “equivalent to” a node in the ontology, or as a “*subtype of*” a *leaf node*.

Comprehensiveness Results

Green are
subtypes

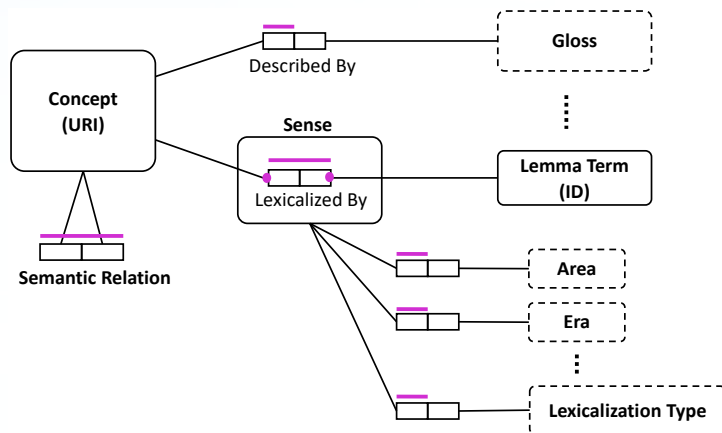
Blue are
equivalents



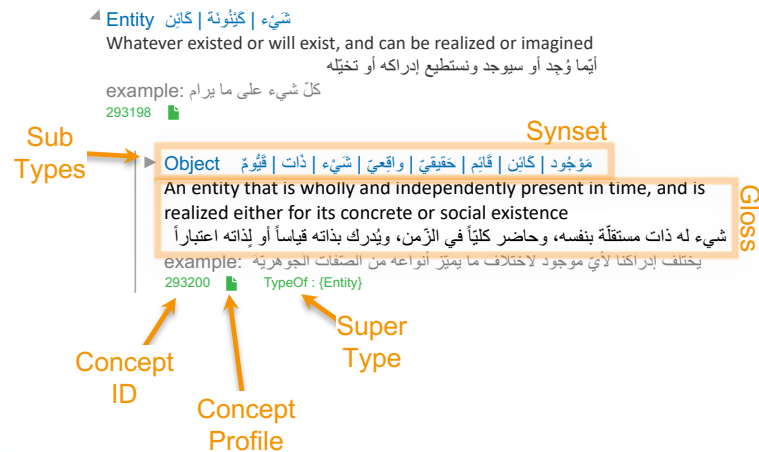
- 1655 (90%) concepts that are correctly placed in the ontology.
- 156 (10%) concepts that are mapped as subclasses of non-leaf nodes illustrate cases of missing top categories in the ontology (we are working on adding them).

Data Representation

Meta Core Data Model



Ontology Portal



→ Also accessible in the RDF W3C Lemon format

URIs Design

To be linked with other resources in **the Linguistic Linked Open Data Cloud**, the URLs should be designed according to the **W3C's Best Practices for Publishing Linked Data**, as the following:

Concepts: each concept is given a URL based on its unique ConceptID:

`http://{domain}/concept/{ConceptID}`

e.g., <http://ontology.birzeit.edu/concept/293254>

Semantic relations: to allow one to retrieve, e.g. the instances, of a given conceptID, the semantic relations for a given concept can be accessed through URLs:

`http://{domain}/concept/{Relation}/{ConceptID}`

e.g., <http://ontology.birzeit.edu/concept/instances/293121>

<http://ontology.birzeit.edu/concept/parts/293121>

Terms: each term is given a URL, which refers to the set of concepts that are lexicalized using a certain term, i.e., that have this *term* among their synsets:

`http://{domain}/concept/{term}`

e.g., <http://ontology.birzeit.edu/concept/virus>

Linking Lexicons with the Arabic Ontology

- ❖ Lexical concepts (in lexicons) are interlinked with the entities in the ontology.
- ❖ Given two entities e_1 and e_2 , a *mapping correspondence* between them is defined as the following:

$$\langle e_1, e_2, R, P, C \rangle$$

- ❖ Progress so far:

Relations	Number of Mappings
SameAs	11400
SubClassOf/SuperClassOf	1050
PartOf/HasPart	100
InstanceOf/Type	770
Similar	125
Total	13445

- In this way, lexical concepts across all lexicons would be semantically linked

Linking Lexicons with the Arabic Ontology

Following the W3C standards:

- ✓ **W3C's Best Practices for Publishing Linked Data**
including the Cool URIs, simplicity, stability, and linking
- ✓ **W3C's RDF Lemon Model**

التسوية levelling | grading
تحريك التربة أثناء إعداد الأرض للري للوصول إلى سطح مستو أو سطح ذي انحدار منتظم.

Hydrology Lexicon ©

```
...  
@prefix aot: <http://ontology.birzeit.edu/term/>.  
@prefix aoc: <http://ontology.birzeit.edu/lexicalconcept/>.  
@prefix aor: <http://ontology.birzeit.edu/lexicon/>.  
  
<aoc:1623> a ontolex:LexicalConcept;  
ontolex:isEvokedBy <aot:Lex-grading>;  
ontolex:isEvokedBy <aot:Lex-levelling>;  
ontolex:isEvokedBy <aot:Lex-تسوية>;  
skos:definition "@ar;تحريك التربة أثناء إعداد الأرض للري للوصول إلى سطح مستو أو سطح ...";  
skos:inScheme <aor:Hydrology_Lexicon_1>.  
  
<aot:lex-grading> a ontolex:LexicalEntry, ontolex:Word;  
ontolex:canonicalForm [ontolex:writtenRep "grading"@en];  
skos:inScheme <aor:Hydrology_Lexicon_1>.  
<aot:lex-levelling> a ontolex:LexicalEntry, ontolex:Word;  
ontolex:canonicalForm [ontolex:writtenRep "levelling"@en];  
skos:inScheme <aor:Hydrology_Lexicon_1>.  
<aot:lex-تسوية> a ontolex:LexicalEntry, ontolex:Word;  
ontolex:canonicalForm [ontolex:writtenRep "تسوية"@ar];  
skos:inScheme <aor:Hydrology_Lexicon_1>.
```

Based On:

Mustafa Jarrar, Hamzeh Amayreh, John McCarae: **Progress on Representing Arabic Lexicons in Lemon**. The 2nd Conference on Language, Data and Knowledge (LDK 2019), Germany. 2019.

Summary

- ❖ New demand to use wordnets as ontologies for managing and retrieving structured data in applications like knowledge graphs.
- ❖ Some issues need to be taken into account when (re-)using wordnet as a linguistic ontology.
- ❖ The Arabic ontology is a linguistic ontology and wordnet at the same time.
- ❖ Following W3C standards is important when building wordnets.

Thank You

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